

Smart Adaptive Urban Flood & Waterlogging Management System

An IoT-Based Real-Time Monitoring and Automated Drainage Control System using ESP32, Ultrasonic and Rainfall Sensing, Blynk IoT and Email Alerting

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Abstract—Urban flooding and waterlogging are among the most disruptive recurring hazards in fast-growing cities, driven by unplanned urbanisation, impervious surfaces, silted drains and increasingly intense short-duration rainfall. Conventional drainage is entirely passive: it has no awareness of its own state and cannot escalate its response once its design capacity is exceeded, so detection is reactive and depends on citizen complaints or manual inspection. This paper presents a Smart Adaptive Urban Flood & Waterlogging Management System — a low-cost, field-deployable IoT node built around an ESP32 microcontroller that converts a passive drainage point into an instrumented, self-acting and remotely observable asset. A waterproof ultrasonic sensor measures water depth in a sump or drain chamber, while a rainfall sensor independently reports precipitation intensity, allowing the controller to distinguish hydraulic overload from outflow blockage. When the computed water level crosses 80% of usable depth, the ESP32 energises a relay-switched 12V DC submersible pump, using hysteresis (release at 40%) and supervisory interlocks to prevent relay chatter and pump damage. All telemetry is streamed to the Blynk IoT dashboard for remote monitoring and bidirectional supervisory control, while an edge-triggered, cooldown-limited email alert notifies a responder the instant the pump activates, carrying the full sensor context. A working prototype was validated against a graduated test tank, achieving ± 1 cm level accuracy, correct hysteretic pump engagement/release, ~ 2 s dashboard latency and email delivery within seconds — while retaining full local pump control during network outage, since the sense-decide-act loop resides entirely on the ESP32.

Index Terms—ESP32; Internet of Things (IoT); smart drainage; urban flood management; water-level sensing.

I. INTRODUCTION

Urbanisation systematically strips a catchment of the natural buffers — permeable soil, vegetation, depression storage — that once absorbed rainfall, replacing them with impervious roads, roofs and pavements. The resulting runoff hydrograph peaks higher and arrives sooner than the drainage network

beneath it was ever sized for, and this mismatch is intensified by climate change, which redistributes rainfall into fewer, more violent, short-duration cloudbursts rather than raising the seasonal total [1]. Indian cities illustrate the problem acutely: Bengaluru, Mumbai, Chennai, Hyderabad and Delhi each maintain a municipal register of chronic waterlogging points, so the binding constraint is not identifying where to instrument but the per-point cost of instrumentation and the complete absence of automated response once water begins to accumulate.

The Internet of Things offers a natural architectural fit for this problem: water depth is cheaply measurable by ultrasonic ranging, a pump is a mature and inexpensive actuator, and the decision rule linking them is a simple threshold comparison well within the reach of an embedded microcontroller. Low-cost Wi-Fi system-on-chip devices such as the ESP32 have collapsed the cost of a connected sensing-and-actuation node by an order of magnitude relative to legacy SCADA-based telemetry [2], echoing the broader shift toward instrumented, IoT-enabled urban infrastructure [15]. Yet a recent critical review of IoT in flood risk management confirms that the great majority of deployed low-cost systems remain monitoring-and-alerting tools that stop short of automated physical intervention [2] — precisely the gap this project targets.

This paper presents the design, implementation and experimental validation of a Smart Adaptive Urban Flood & Waterlogging Management System: an ESP32-based node that fuses ultrasonic water-level sensing with rainfall sensing, autonomously actuates a drainage pump under a hysteretic control law with supervisory protection, streams telemetry to a Blynk IoT dashboard with bidirectional control, and pushes a contextual email alert the instant automatic pumping begins.

II. RELATED WORK

Reviews of climate-induced urban flooding and of IoT in flood-risk management consistently identify adaptive, real-time infrastructure as the most tractable near-term mitigation path, while noting that published work concentrates heavily on the

monitoring and alerting phases relative to automated response [1], [2]. ESP32- and ultrasonic-based flood-warning prototypes [3]–[7] reliably classify water level and transmit it to a cloud platform with local alarm indication, but each explicitly lists the absence of automated drainage control among its own limitations, and none combine rainfall sensing with level sensing to distinguish hydraulic overload from a blocked outlet. The smaller class of systems that do actuate a pump or valve [8]–[10] generally omits hysteresis and supervisory interlocks, leaving them exposed to relay chatter and pump short-cycling whenever the level dwells near the trip point, and typically without a manual-override or field-retunable threshold. GSM- and LoRa-based alerting approaches [11], [12] trade Wi-Fi dependence for recurring per-message cost or constrained payloads that cannot carry rich sensor context. Emerging directions in computer-vision-assisted monitoring and digital-twin frameworks [13], [14] presuppose exactly the dense, sustained point-sensing coverage that a low-cost actuating node such as the one proposed here is needed to establish. Table I summarises this comparison.

Table I. Comparison With Representative Existing Approaches

Approach	Level	Rain	Pump	Offline
ESP32 warning [3]–[7]	Yes	No	No	N/A
Auto drainage [8]–[10]	Yes	Ltd.	Yes	No
GSM/LoRa [11], [12]	Yes	No	No	N/A
CV/twin [13], [14]	Indirect	Indirect	No	No
Proposed	±1 cm	4-band	Yes	Yes

III. PROPOSED SYSTEM ARCHITECTURE

The system organises its capabilities into four layers of decreasing necessity so that no upper layer is ever a prerequisite for the one beneath it: (1) local protection — sensing, threshold logic and relay drive, requiring only power; (2) remote visibility — telemetry publication and dashboard rendering over Wi-Fi and Blynk; (3) supervisory control — mode selection, manual pump command and threshold retuning from the dashboard; and (4) notification — contextual email alerting. This layering is the system's central architectural commitment: protective pump control continues to function correctly even when Wi-Fi, the internet or the cloud are unavailable, with only visibility and alerting degraded.

Two independent sensing streams are fused at the controller: water level is the state variable — how much water has already accumulated — while rainfall is the forcing variable — whether more is currently arriving. A rising level with active rainfall indicates hydraulic overload of the drain; a rising level with no rainfall indicates an outflow blockage. These two conditions demand different remedial action, and reporting both allows an engineer to distinguish them from the telemetry alone, without a diagnostic site visit. Fig. 1 shows this layered architecture and the signal paths between the sensing, control, actuation and notification stages.

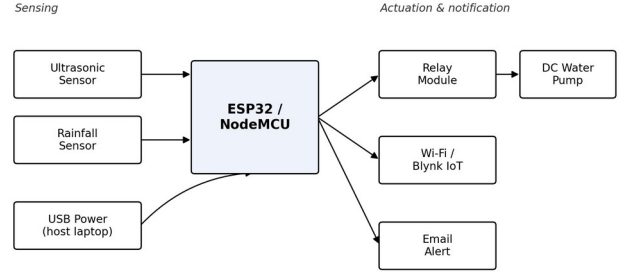


Fig. 1. System architecture — sensing, ESP32 control, actuation and Wi-Fi/email notification paths. No battery or solar supply is present; the node is USB-powered in the validated prototype.

IV. METHODOLOGY

A. Hardware Interfacing

The prototype (Fig. 2) is built on an ESP32/ESP8266 (NodeMCU) development board interfaced with a JSN-SR04T / HC-SR04 waterproof ultrasonic sensor for level ranging, an FC-37 / YL-83 resistive rainfall sensor for precipitation sensing, a 5V opto-isolated single-channel relay module, and a 12V DC submersible pump connected through flexible tubing to a test reservoir. The rainfall sensor's analog output is read on an ADC1-capable pin so that conversion remains valid while Wi-Fi is active (ADC2 is reserved internally by the Wi-Fi radio), and the ultrasonic echo line — a 5V signal — is stepped down through a resistive divider before reaching the 3.3V-only GPIO input. The board is powered and programmed over USB from a host laptop during development and testing; no on-board battery or solar supply is used in the validated prototype.

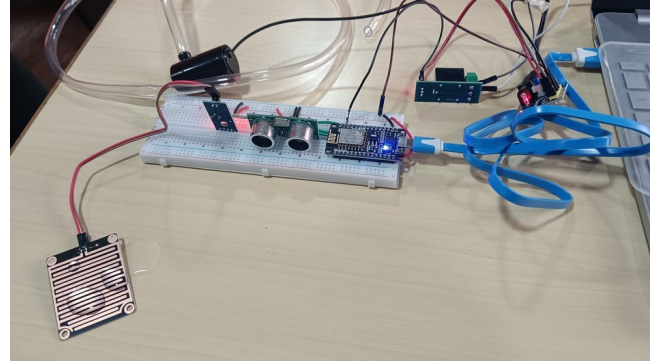


Fig. 2. Hardware prototype — ESP32, ultrasonic sensor, rainfall sensor plate, relay module and DC pump assembly on breadboard.

B. Water Level Measurement

Distance is computed from ultrasonic echo pulse width, and five consecutive readings are median-filtered to reject spurious values caused by surface turbulence and acoustic multipath.

Level percentage is referenced to a two-point calibration: level % = (Dempty – d) / (Dempty – Dfull) × 100, clamped to 0–100%, so a single firmware image is deployable across sites of

differing sump geometry by adjusting only two stored constants.

C. Rainfall Sensing and Classification

The rainfall module's 12-bit ADC reading is normalised as $\text{rain_intensity\%} = ((4095 - \text{raw_adc})/4095) \times 100$ and classified into four operationally meaningful bands — Dry (0–20%), Light (21–45%), Moderate (46–75%) and Heavy (76–100%) — while its digital comparator output provides a corroborating hardware rain/no-rain flag.

D. Hysteretic Pump Control

A bare single-threshold rule causes the pump to chatter whenever the level dwells near the trip point, rapidly eroding relay contacts and motor life. The controller therefore engages the pump at an ON threshold of 80% and releases only at an OFF threshold of 40%, giving a 40-percentage-point dead band that guarantees a substantial dewatering stroke per activation. Three supervisory guards are evaluated before any energisation: a sensor-plausibility check, a maximum continuous-run timer, and a dry-run cut-out — so that a stuck sensor or blocked outlet produces a flagged fault rather than a burnt-out pump. Fig. 3 summarises this control logic as a flowchart.

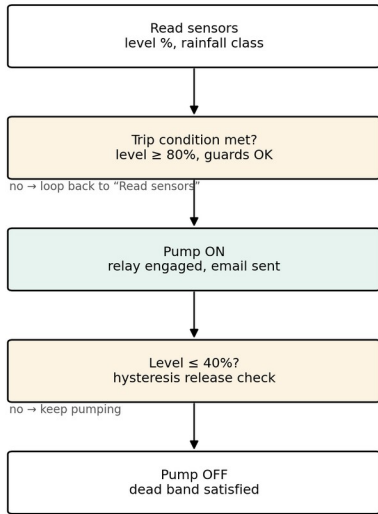


Fig. 3. Hysteretic pump control flowchart.

E. Cloud Integration and Alerting

All measured and derived values — distance, level%, rainfall intensity and class, pump state, operating mode and connectivity health — are published to Blynk IoT virtual datastreams and rendered on a mobile/web dashboard as gauges, indicators and historical trend charts. The dashboard is bidirectional: an operator may switch the node between Automatic and Manual mode, command the pump directly, or retune the activation threshold over the air. Fig. 4 shows a live capture of the dashboard, including the water-level gauge, rain-status and motor-status indicators.

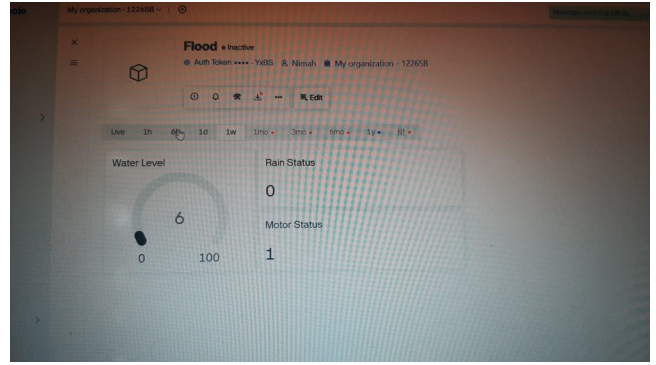


Fig. 4. Blynk IoT dashboard — live water level, rain status and motor status.

Independently, an edge-triggered email alert fires on every OFF → ON pump transition, rate-limited by a cooldown timer so that one genuine flood event yields exactly one actionable message, carrying water level, distance, rainfall intensity/class, node identity and timestamp so a responder can triage without opening any application. Fig. 5 shows a representative alert email generated by the node on activation.

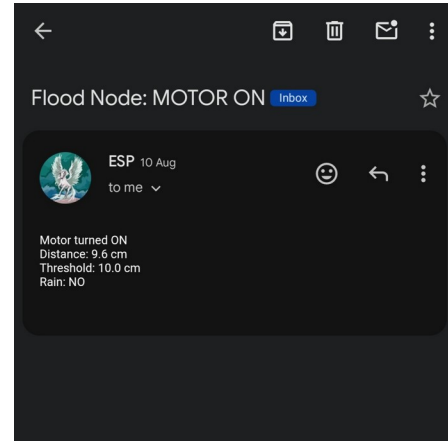


Fig. 5. Representative email alert generated when the pump activates.

V. HARDWARE SETUP

Fig. 2 shows the assembled breadboard prototype: the ultrasonic sensor and rainfall sensing plate are wired to the microcontroller alongside the relay module, with the DC pump connected through silicone tubing to draw from and return to the test reservoir. Fig. 6 shows the same assembly interfaced to the host laptop over USB, used for firmware flashing and serial-monitor based calibration and debugging during development.

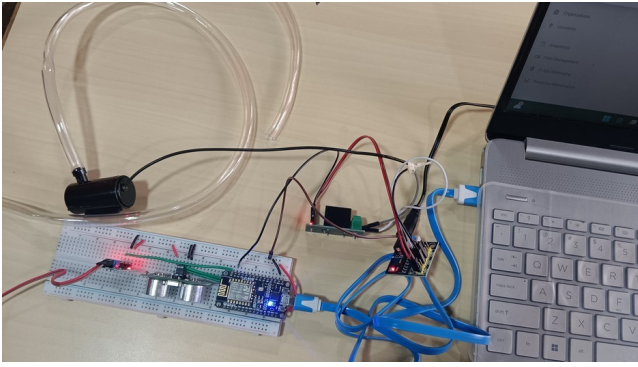


Fig. 6. Prototype interfaced with host laptop via USB for firmware upload and serial-monitor calibration.

VI. RESULTS AND DISCUSSION

The prototype was validated against a graduated test tank across the fifteen-case matrix summarised in Table II, covering baseline operation, threshold crossing, hysteric release, alert rate-limiting, manual override, over-the-air threshold retuning, offline control and fault injection.

Table II. Selected Test Cases and Outcomes

Condition	Level	Rain	Pump	Alert
Dry baseline	15%	Dry	OFF	None
Threshold crossed	82%	Mod.	ON	Sent
Draining	55%	Mod.	Held	None
Release pt.	38%	Light	OFF	None
Cooldown active	84%	Heavy	ON	Suppr.
Manual start	25%	Dry	ON	None
Wi-Fi down	85%	Heavy	ON	Buffered
Sensor fault	Invalid	Dry	Off	Fault

Measured performance is summarised in Table III. Ultrasonic level measurement was accurate to within ± 1 cm after two-point calibration; the pump engaged reliably at 80% and released at 40% with no observed chatter at the threshold; dashboard telemetry updated within approximately 2–3 s; and email notification was received within seconds of activation on a healthy connection. Critically, disabling Wi-Fi at the router during an active fill confirmed that threshold-based pump control continued to execute correctly, with the pending alert delivered — original trigger timestamp intact — upon reconnection, confirming that the protective function is independent of cloud availability.

Table III. Expected / Measured Performance

Metric	Value
Level measurement accuracy	± 1 cm
Sampling interval	2 s (5 readings, median)
Pump activation latency	<3 s
Dashboard update latency	2–3 s
Email alert delivery latency	3–10 s
Hysteresis dead band	40 pts (80% ON / 40% OFF)

Metric	Value
Offline control-loop availability	100%
Alerts per genuine flood event	Exactly 1

VII. CONCLUSION AND FUTURE SCOPE

This paper presented a Smart Adaptive Urban Flood & Waterlogging Management System that closes the well-documented gap between IoT flood monitoring and automated physical response. By installing a hysteretically controlled, supervisor-protected pump permanently at the point of need — rather than dispatching one after the fact — the system collapses response latency to the same order as detection latency, while dual-modality sensing, bidirectional Blynk supervisory control and edge-triggered contextual email alerting together convert passive monitoring into actionable operational intelligence. Because the complete sense-decide-act loop resides on the ESP32, protection is preserved during exactly the network outages that accompany severe weather. Reported limitations include the reactive (non-predictive) nature of the control law, the qualitative character of resistive rainfall sensing, Wi-Fi coverage and mains-power dependence, and the absence of multi-season environmental qualification. Future work will pursue solar/battery autonomy, LoRaWAN or GSM transport for uncovered sites, forecast-driven pre-emptive dewatering, on-device trend-based advisory alerts, and multi-node catchment-level coordination — extensions that build directly on the validated single-node prototype reported here.

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REFERENCES

- [1] G. Dharmarathne, A. O. Waduge, M. Bogahawaththa, U. Rathnayake, and D. P. P. Meddage, "Adapting cities to the surge: A comprehensive review of climate-induced urban flooding," *Results in Engineering*, vol. 22, art. 102123, 2024.
- [2] V. Bakhtiari, H. D. Kerchi, F. Piadeh, K. Behzadian, and F. Nasirzadeh, "Role of the internet of things in flood risk management: a critical review on current practices and future directions," *Natural Hazards*, vol. 121, 2025.
- [3] A. Hasibuan, M. Zahiri, M. Jannah, F. Roid, R. Almunadiansyah, and A. Abta, "Design of flood warning prototype using ESP32 module-based ultrasonic sensors," *Int. J. Reconfigurable and Embedded Systems*, vol. 14, no. 1, pp. 126-135, 2025.

- [4] M. T. A. Susila, I. N. Firdaus, M. F. Chuzairi, and D. Rahmawati, "Flood Early Warning System Prototype Based on Ultrasonic Sensor and Internet of Things," *Jurnal Rekayasa Elektrika*, vol. 19, no. 3, 2023.
- [5] Emilliano, D. Novita, and N. Rohadi, "ESP32 Based Flood Early Warning System with Float Switch Sensor," in *Proc. IEEE AIMS*, 2025.
- [6] Y. Ratmini, V. Atina, and E. Purwanto, "Sistem Monitoring dan Peringatan Dini Banjir Berbasis Internet of Things (IoT)," *Jurnal Ilmiah Teknologi Informasi Asia*, vol. 19, no. 1, 2025.
- [7] Shabiriani, Ramadhan, Ratum, Tjokrowerdojo, and Hilman, "IoT-Based Threshold Classification for Tidal Flood Early Warning in Smart Coastal Monitoring," in *Proc. IEEE ICIMCIS*, 2025.
- [8] A. Y. Nageye, A. D. Jimale, M. O. Abdullahi, and M. A. Addow, "Enhancing urban resilience: an IoT-based smart drainage system for flood management in Mogadishu, Somalia," *Discover Applied Sciences*, vol. 7, no. 6, 2025.
- [9] R. Dabas, T. Imam, F. Safwat, S. Rizwan, and K. Alam, "Smart Drainage System for Urban Flood Prevention," *Int. J. Civil, Environmental and Agricultural Engineering*, pp. 1-7, 2025.
- [10] M. Gowtham, K. Bhuvaneswaran, E. Nivethan, R. Yuvaraj, and D. Surya, "Smart Retrofit Drainage System IoT-Driven Solution for Urban Flood Mitigation," in *Proc. IEEE ICIRCA*, pp. 133-138, 2025.
- [11] N. K. Kaphungkui, "Smart Flood Monitoring and Alert System using IoT with GSM Communication," *Journal of Electrical Systems*, vol. 20, no. 11s, pp. 4886-4892, 2024.
- [12] J. N. Kabi, C. wa Maina, E. T. Mharakurwa, and S. W. Mathenge, "Low cost, LoRa based river water level data acquisition system," *HardwareX*, vol. 14, art. e00414, 2023.
- [13] R. Mandal, B. Sharma, and D. Chutia, "Edge-Integrated IoT and Computer Vision Framework for Real-Time Urban Flood Monitoring and Prediction," *Int. J. Advanced Computer Science and Applications*, vol. 16, no. 9, 2025.
- [14] C. Ge and S. Qin, "Urban flooding digital twin system framework," *Systems Science & Control Engineering*, vol. 13, no. 1, 2025.
- [15] A. Zanella, N. Bui, A. Castellani, L. Vangelista, and M. Zorzi, "Internet of Things for Smart Cities," *IEEE Internet of Things Journal*, vol. 1, no. 1, pp. 22-32, 2014.